

Independent Study Final Report

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Section 1: Introduction

The study focused on investigating the impact of wildfire elements, specifically CIPP pipes and metal slags, on the contamination process. The primary motivation for this research was to better understand how these elements contribute to contamination and to determine whether their presence significantly affects the environment. The goal was to gain insights into the contamination process, identify potential risks, and provide evidence to confirm or refute the existence of contamination caused by these materials.

Section 2: Methodology

We began the study with a hypothesis that hydrogen sulfide (H_2S) would be released during the heating of metal slags. To test this, we conducted experiments by heating the metal slags to a temperature where H_2S release was expected and used a sulfur meter to measure the emissions. When no significant readings were observed, we shifted our focus to analyzing the total organic carbon content, hypothesizing that the slags, which are blasted with coke, might contain measurable levels of organic carbon.

In the second phase, we conducted leachate testing by resting the slag inside the measured water. The tests were carried out over intervals of 24 hours, 3 days, and 1 week. For each time frame, we collected five samples with varying dilutions: 10%, 20%, 50%, 90%, and full-strength leachate. These samples were analyzed to create a calibration curve using a TOC analyzer, providing a detailed assessment of the slag leachate in the water. This procedure was repeated for all three major samples, ensuring consistency and reliability in the results.

Section 3: Key Findings

The study revealed a correlation between the slag leachate and an increase in total organic carbon levels. While the percentage increase was not exceptionally high, it indicated a trend worth further investigation. Using basic statistical measures and standardization, we analyzed the data and conducted a t-test to evaluate the significance of the results. The t-test confirmed the hypothesis, providing evidence of a correlation between the slag leachate and the observed changes in total organic carbon. This finding highlights the potential environmental impact of slag leachate and offers a basis for further research.

Section 4: Challenges

One of the primary challenges of this study was the lack of existing research on the specific impact of metal slags and CIPP pipes on contamination processes. This gap in knowledge made it difficult to establish a strong foundation or compare findings with prior studies. Additionally, further testing is required to better understand the specific types of organic carbon present in the leachate and whether they pose a potential threat to the environment. Questions remain about the chemical composition of these organic compounds, their reactivity, and their long-term environmental impact.

Section 4: Conclusion

This study confirmed a correlation between slag leachate and an increase in total organic carbon, highlighting potential environmental concerns. Further research is necessary to identify specific organic compounds, assess their environmental impact, and explore mitigation strategies.